

Figure 4: Output of *ofswitch13-modify.cc*

Figure 5: Output of the log file

In the program's code, to view the log file, set `trace=true`. This will generate `switch-stats-1.log` in the `ns-dev` folder.

Network Animator, also known as NetAnim, is used to graphically portray projects in ns-3. It is an offline animator, which animates the XML file generated during the simulation program in ns-3. ns-2 has many default animators for use but ns-3 is furnished with no default animator. So, we have to integrate NetAnim with ns-3. NetAnim version 3.107 is used for visualising.

Add the following few lines of code:

The above four lines of code will set the position of the hosts (nodes) at the given coordinates on the X-Y plane (refer to the screenshot in Figure 6), and then generate the *ofsI3-modifv.xml* file for the *ofswitchI3-modifv.cc*.

The other three nodes are the created hosts.

- Node 0: 10.97.7.1
- Node 1: 10.97.7.2
- Node 2: 10.97.7.3

Figure 7 is a screenshot of the generated XML file with graphical simulation in NetAnim.

- [1] <https://www.nsnam.org/docs/release/3.26/doxygen/index.html>
- [2] <https://www.opennetworking.org/images/stories/downloads/sdn-resources/onf-specifications/openflow/openflow-switch-v1.5.0.noipr.pdf>
- [3] <http://www.lrc.ic.unicaup.br/ofswitch13/ofswitch13.pdf>

The source repository can be downloaded from:
https://bitbucket.org/vashsquare/ns3_support

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